Charotar University of Science and Technology

M.Sc. (Nano Science & Technology) Semester II Examination May 2011

NT710 Nano Characterization Techniques

Date 5th May 2011 Time: 10:00 to 10:30 Max. Marks 20

Part A

(Do as directed)

Q1. Show that the atomic packing factor for fcc structure is 0.74. (1)

Q2. The resolution of X-ray is ______________________________. (1)

Q3. For a given hcp structure, A= 0.1113 and typical $\sin^2\theta = 0.4698$. Determine c/a ratio. (1)

Q4. Name different modes of Scanning Electron Microscopy. (1)

Q5. ____________________________________ mode of SEM is commonly used to reveal the surface morphology of the specimen. (1)

Q6. Name different modes of Transmission Electron Microscopy. (1)

Q7. In the dynamic mode of AFM, the cantilever is driven at ____________________. (1)

(i) At lower frequency (ii) At resonance frequency

(iii) At higher frequency

Q8. STM measures ________________________ between tip atoms and surface atoms of specimen while AFM measures ________________________. (1)

Q9. Wavelength range of UV radiation is ______________, visible radiation is __________ __________ and NIR radiation is ____________________. (1)

Q10. Write the Beer-Lambert equation and define each symbol (with suitable unit). (1)

Q11. Polar solvent ________________________ the energy of $\pi \rightarrow \pi^*$ transition. (1)

(i) decreases (ii) increases

(iii) does not affect

Q12. A shift of an absorption maximum towards longer wavelength is called ______________ __________ while that towards shorter wavelength is called ______________________. (1)

(i) Bathochromic shift (ii) Hypsochromatic shift

(iii) Hypochromatic shift (iv) Hyperchromatic shift

Q13. __________________________ recombination of Photoluminescence process can be used to estimate the material bandgap (E_g). (1)

(i) Band to Band (ii) Free to bound

(iii) Donor-Acceptor (iv) Exciton –Phonon

Q14. Band to band recombination dominates at ______________________________ while free to bound transition dominates at ____. (1)

(i) room temperature (ii) high temperature

(iii) low temperature

Q15. Energy of a particular characteristic X-ray __________________ with increasing atomic number. (1)

(i) increases (ii) decreases

(iii) remains constant

Q16. In XPS technique ____________________________ is being probed on specimen. (1)

(i) Laser beam (ii) visible radiation

(iii) X-ray beam (iv) electron beam

Q17. AES cannot detect the presence of __________________ and ___________________ atoms. (1)

Q18. In the case of $KL_{2,3}L_{2,3}$ Auger transition, Auger electron will be emitted from ________ shell. (1)

Q19. Thermogravimetry provides the information about ___________________involved in the reaction while Differential Scanning Calorimetry gives information about _____________________.(1)

(i) volume change (ii) mass change

(iii) colour change (iv) heat exchange

Q20. ________________ transitions are IR active, while __________________ are Raman active. (1)

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Part B

(Write each section on separate answer book)

Section I

Q1. (a) Explain: structure factor, polarization factor, multiplicity factor. Why do we calculate related integrated intensities? **(4)**

(b) Index the pattern and determine the lattice parameter for following 2theta values. Use common quotient and analytical methods.
2theta = 44.53, 51.89, 76.45, 93.01, 98.51, 122.12 **(6)**

OR

Q1. (a) The powder pattern of Al made with CuK_α radiation contains ten lines whose $\sin^2\theta$ values are 0.1118, 0.1487, 0.294, 0.403, 0.439, 0.583, 0.691, 0.727, 0.872 and 0.981. Index these lines and calculate lattice parameters. **(6)**

(b) A pattern is made of a cubic substance with unfiltered chromium radiation. The observed $\sin^2\theta$ values ad intensities are 0.265 (m), 0.321 (vs), 0.528(w), 0.638 (s), 0.793 (s) and 0.958 (vs). Index these lines and state which are due to K_α and which due to K_β radiation. Determine the bravias lattice and lattice parameter. **(4)**

Q2. (a) Give a brief description of different operational modes of Transmission Electron Microscopy. **(4)**

OR

(a) Enumerate differences between Scanning Tunneling Microscopy and Atomic Force Microscopy. (4)

(b) How the particle size is being estimated using Dynamic Light Scattering? **(6)**

Section II

Q3. (a) What are the different stages of a photoluminescence process? Also explain different possible recombination mechanisms. **(6)**

(b) Shortly describe different parts of a typical UV-VIS spectrometer. **(4)**

OR

(b) Generally sets of absorption frequencies observed in the spectrum of any material are less than calculated number of vibrational modes. Justify this statement. **(4)**

Q4. (a) Explain how AES can be used to obtain elemental information. Also describe the chemical information, which can be obtained from AES. **(6)**

(b) Explain how the characteristics x-rays are generated form a material. **(4)**

OR

(b) What is the final state effect in x-ray photoelectron spectroscopy? **(4)**

Q5. (a) List out different applications of Raman spectroscopy? **(2)**

(b) Explain the principle of Thermogravimetric analysis. **(4)**

OR

(b) Explain the principle of Differential Scanning Calorimetry. **(4)**

(c) Describe the construction of a typical Raman spectrometer. **(4)**

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